

Phenom ParticleX GSR Desktop SEM

Dedicated gunshot residue analysis on a desktop SEM





GSR particles are carried in the cloud of smoke after a gun discharge and deposited on the shooter's hand and surroundings.

The Thermo Scientific™ Phenom™ ParticleX GSR Desktop SEM is the first dedicated desktop SEM that can run automated GSR analysis. The full integration of hardware and software brings a unique user-friendly and reliable solution to any forensic lab.

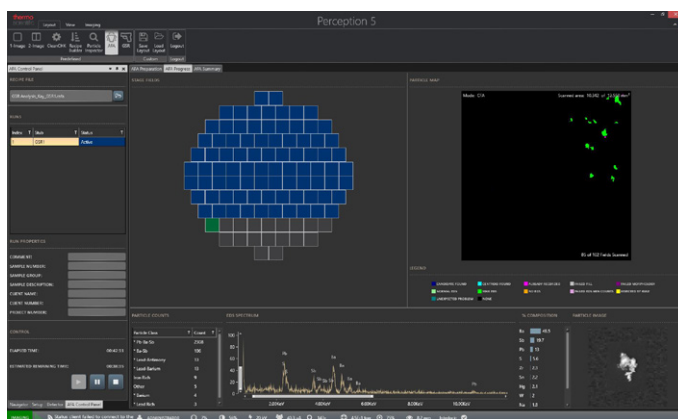
Introduction

Gunshot residue (GSR) analysis plays an important role in the determination of when a firearm has been used in a crime. Established GSR analysis techniques are based on the use of a scanning electron microscope (SEM), which is used to scan the sample and find suspect GSR particles. If a suspect particle is found, energy dispersive spectroscopy (EDS) is used to identify the chemical composition of that particle. Most common search criteria are the presence of Pb, Sb, and Ba. However, detection of Pb-free primers, such as Ti and Zn, is a requirement as well. The Phenom Perception GSR Desktop SEM is equipped with a CeB₆ source. A CeB₆ source ensures very stable beam current compared to the traditional tungsten sources, while remaining affordable compared to a FEG source.

With a typical source lifetime of >1,500 hours, the Phenom ParticleX GSR Desktop SEM is ideal for usability and uptime. On top of that, the CeB₆ has a gradual degradation at the end-of-life. This makes the exchange of the source easy to plan, and no automated runs will be interrupted due to broken filaments.

High throughput, reliable results

Thanks to the fully motorized stage, the Phenom ParticleX GSR Desktop SEM can handle a scan area of 100x100 mm. The software uses the internal scan control of the SEM. This enables more accurate beam positioning, which especially helps when revisiting the particle in the GSR verification phase. A standard GSR sample holder can hold 30 12 mm GSR pin stubs plus the necessary calibration samples.



View during live gunshot residue analysis.

Perception software

Automated gunshot residue analysis

Intuitive user interface limits training time

Run your desktop SEM day and night to enhance efficiency

Extensive reporting options that support manual revisiting and validation of particles

Compliant with ASTM E1588 and ENFSI Best Practice Guidelines

Typically >98% hit rate on artificial Plano GSR sample

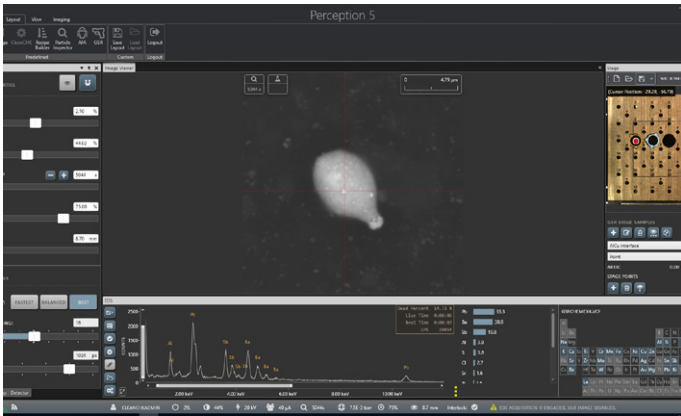
Reclassification and requantification of results

High-speed analysis while maintaining accurate and repeatable results

Versatile output formats: PDF, Word, Excel, CSV

Perception GSR Software works based on recipes that are easy to custom build from standard building blocks. Once a standard operating procedure (SOP) is determined for your lab, it is easy to execute pre-set recipes for different cases. The stage layout in combination with the navigation camera of the Phenom desktop SEM makes it fast to adjust runs for multiple samples. Fully automated particle location and EDS analysis will give repeatable results. Revisiting is intuitive with the high-accuracy stage and will automatically generate detailed confirmation reports.

This desktop SEM can also be used for many other forensics applications, such as ballistics, paint analysis and fiber characterization. Moreover, the Phenom ParticleX GSR Desktop SEM is easy to set up and transport and can be relocated without difficulty. The system does not require any special facilities or components, such as compressed air, chillers, liquid nitrogen, EM shielding or cooling water. Additionally, it has a low CO₂ footprint (average energy usage of 163 Watts).



Gunshot residue particle with optical overview and stage layout.

Fully integrated EDS

The dedicated software package Element Identification (EID) is used to control the fully integrated EDS detector. This EID software is standard as part of the Phenom ParticleX GSR Desktop SEM. Analysis has become as easy as imaging, since there is no need to switch between external software packages or computers. The CeB6 electron source in the Phenom desktop SEM is used to generate the highest X-ray count rate in its market segment, allowing fast and accurate results.

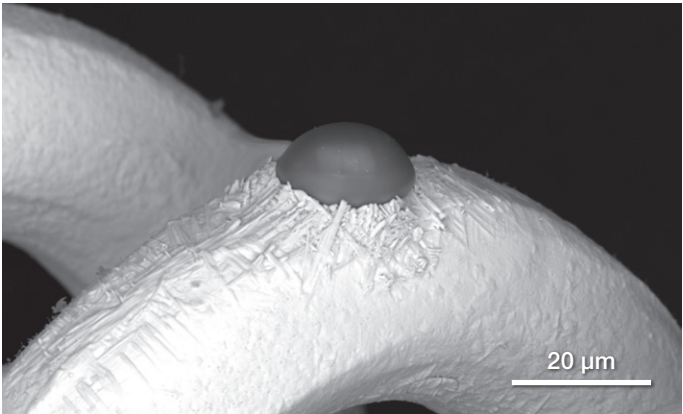
The EID software package allows you to identify nearly all materials in the periodic table, starting from boron (5) and ranging up to americium (95). It is a perfect analysis tool for a wide range of samples and applications. Projects can be stored locally or on the network, where they can be analyzed at a later stage or offline.

The EID software package runs smart algorithms with advanced peak analysis to optimize the auto-identification functionality, while still allowing for manual adjustments at any time in the analysis process. The intuitive step-by-step process within the software helps you collect all X-ray results in an organized and structured way.



Sample holder with up to 36 12 mm pin stubs.

Imaging	
Detection modes	
Light	Magnification range: 3–16x
Electron	Magnification range: 160–200,000x
Illumination	
Light	Bright field / dark field modes
Electron	- Long lifetime thermionic source (CeB₆) - Multiple beam currents
Acceleration voltages	- Default: 5 kV, 10 kV and 15 kV - Advanced mode: adjustable range between 4.8 kV and 20.5 kV imaging and analysis mode
Vacuum levels	Low - medium - high
Resolution	<10 nm
Detector	
Standard	Backscattered electron detector, and energy dispersive spectroscopy detector
Optional	Secondary electron detector
Digital image detection	
Light optical	Proprietary high-resolution color navigation camera, single-shot
Electron optical	High-sensitivity backscattered electron detector (compositional and topographical modes)
Image formats	
JPEG, TIFF, PNG	
Image resolution options	
960 x 600, 1920 x 1200, 3840 x 2400 and 7680 x 4800 pixels	
Data storage	
USB flash drive, Network, workstation with SSD	
Sample stage	
Computer-controlled motorized X and Y	

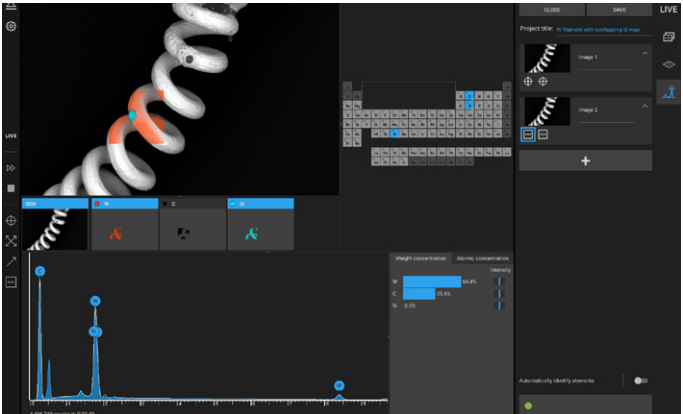


Tungsten filament with glass droplet.

EDS	
Detector type	- Silicon Drift Detector (SDD) - Thermoelectrically cooled (LN₂ free)
Detector active area	25 mm ² or 70 mm ²
X-ray window	Ultra thin silicon nitride (Si ₃ N ₄) window allowing detection of elements B to Am
Energy resolution	Mn Kα ≤132 eV
Processing capabilities	Multi-channel analyzer with 2048 channels at 10 eV/ch
Max. input count rate	300,000 cps
Hardware integration	Fully embedded

Software	
- Integrated in Phenom user interface - Integrated column and stage control - Auto-peak ID - Iterative strip peak deconvolution - Confidence of analysis indicator - Export functions: CSV, JPG, TIFF, ELID, EMSA	

Report	
DOCX format	



EDS map on a tungsten filament with a molten glass droplet.

Elemental Mapping & Line Scan	
Elemental mapping	
Element selection	Individual user-specified maps, plus backscatter image and miximage
Backscatter image and mix-range	
Selected area	Any size, rectangular
Mapping resolution range	32 x 20 to 960 x 600 pixels
Pixel dwell time range	1–500 ms
Line scan	
Line scan resolution range	16–512 pixels
Line scan dwell time range	10–500 ms
Report	
DOCX format	
SED	
Detector type	Everhart Thornley

System

Dimensions & weight

Imaging module	316(w) x 587(d) x 625(h) mm, 75 kg
Diaphragm vacuum pump	145(w) x 220(d) x 213(h) mm, 4.5 kg
Power supply	260(w) x 260(d) x 85(h) mm, 2.3 kg
Monitor (24")	531(w) x 180(d) x 511(h) mm, 5.6 kg
Workstation	- Powerful workstation including SSD storage 93(w) x 293(d) x 290(h) mm, 5.6 kg

Sample size

- Max. 100 mm x 100 mm (up to 36 x 12 mm pin stubs)
- Max. 40 mm height (optional up to 65 mm)

Scan area

100 mm x 100 mm

Sample loading time

Light optical	<5 s
Electron optical	<60 s

Site requirements

Ambient conditions

Temperature	15°C ~ 30°C (59°F ~ 86°F)
Humidity	Between 20% and 80% RH
Power	Single phase AC 100–240 Volt, 50/60 Hz, 163 W average, 348 W max

Recommended table size

150 x 75 cm, load rating of 150 kg

Sample holders and inserts

- Manual-Z sample holder
- Motorized-Z sample holder (optional)
- 49-stub insert (optional)

 Learn more at thermofisher.com/phenom-particle-x-gsr

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